Т3. Случ. вел. имеет экп. распределение:

$$p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \theta > 0$$

По выборке объема $n = 3$ найдены оценки параметра θ: $\tilde{\theta}_1 = \bar{X}$, $\tilde{\theta}_2 = X_{(2)}$ (2-ой член вар. ряда)

а) Проверить оценки на несмещенность. Исправить эти оценки, если необходимо.

в) Какая из исправленных оценок более эффективна?

с) Исследовать эти оценки на эффективность с помощью нер-ва Крамера-Рао.

а) $\tilde{\theta}_1 = \bar{X}$, $M\tilde{\theta}_1 = M\left[\frac{1}{n}\sum_{k=1}^{n} X_k\right] =$

$$= M\xi = \int_0^{+\infty} \frac{x e^{-x/\theta}}{\theta} dx = -\frac{\theta}{\theta}\int_0^{+\infty} x \, de^{-x/\theta} =$$

$$= -\left[x e^{-x/\theta}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-x/\theta} dx\right] = \int_0^{+\infty} e^{-x/\theta} dx =$$

$$= -\theta e^{-x/\theta}\Big|_0^{+\infty} = \theta \Rightarrow \tilde{\theta}_1 \text{ - несмещенная}$$

$\tilde{\theta}_2 = X_{(2)}$. Т.к. $\xi \sim F(x)$, то $X_{(2)} \sim \varphi(t) =$

$$= \sum_{i=2}^{n} C_n^i F^i(t)(1-F(t))^{n-i}, \quad F(x) = \int_{-\infty}^{x} p(t)dt =$$

$$= \begin{cases} 0, & x \leq 0 \\ \int_0^x e^{-t/\theta}/\theta \, dt = -e^{-t/\theta}\Big|_0^x = 1 - e^{-x/\theta}, & x > 0 \end{cases}$$

$$\varphi(t) = C_3^2 (1-e^{-t/\theta})^2 e^{-t/\theta} + C_3^3 (1-e^{-t/\theta})^3 =$$

$$= 3\left(1-e^{-t/\theta}\right)^2 e^{-t/\theta} + \left(1-e^{-t/\theta}\right)^3 = \left(1-e^{-t/\theta}\right)^2\left(3e^{-t/\theta}+1-e^{-t/\theta}\right) =$$

$$= \left(1-e^{-t/\theta}\right)^2\left(1+2e^{-t/\theta}\right)$$

$$g(t) = \varphi'(t) = 2\left(1-e^{-t/\theta}\right)\frac{1}{\theta}e^{-t/\theta}\left(1+2e^{-t/\theta}\right) +$$

$$+\left(1-e^{-t/\theta}\right)^2 \cdot \frac{2}{\theta}e^{-t/\theta} =$$

$$= \frac{2}{\theta}e^{-t/\theta}\left(1-e^{-t/\theta}\right)\left(1+2e^{-t/\theta}-1+e^{-t/\theta}\right) =$$

$$= \frac{6}{\theta}e^{-2t/\theta}\left(1-e^{-t/\theta}\right) \quad \{(0;+\infty)\}$$

$$M X_{(2)} = \frac{6}{\theta}\int_0^\infty e^{-2x/\theta}\cdot x\left(1-e^{-x/\theta}\right)dx =$$

$$= \frac{6}{\theta}\left[\int_0^\infty x e^{-2x/\theta}dx - \int_0^\infty x e^{-3x/\theta}dx\right] =$$

$$= \frac{6}{\theta}\left[I_2 + \left(-I_3\right)\right] = \frac{6}{\theta}\left[\frac{\theta^2}{4} - \frac{\theta^2}{6}\right] = \frac{6}{\theta}\left[\frac{5\theta^2}{36}\right] \ominus$$

$$\left\{ \begin{aligned} &I_\alpha = \int_0^\infty x e^{-\frac{\alpha x}{\theta}}dx = -\frac{\theta}{\alpha}\int_0^\infty x\,de^{-\frac{\alpha x}{\theta}} = \\ &\alpha>0 \\ &= -\frac{\theta}{\alpha}\left[x e^{-\frac{\alpha x}{\theta}}\Big|_0^\infty - \int_0^\infty e^{-\frac{\alpha x}{\theta}}dx\right] = \frac{\theta}{\alpha}\int_0^\infty e^{-\frac{\alpha x}{\theta}}dx = \\ &= \frac{\theta}{\alpha}\left(-\frac{\theta}{\alpha}e^{-\frac{\alpha}{\theta}x}\Big|_0^\infty\right) = \frac{\theta^2}{\alpha^2} \end{aligned} \right\}$$

$\ominus$ $\frac{5\theta}{6}$ – смещенная; $\tilde{\tilde{\theta}}_2 = \frac{6}{5}X_{(2)}$ – исправленная

в) Исправленная оценка только одна

Сравним на эффективность: $\tilde{\theta}_1 = \bar{X}$ и $\tilde{\tilde{\theta}}_2 = \frac{6}{5}X_{(2)}$

Вычислим дисперсии:

$$D\hat{\theta}_1 = D\bar{x} = \frac{1}{n^2} n D\xi = \frac{1}{n} D\xi$$

$$M\xi^2 = \int_0^\infty x^2 e^{-x/\theta}/\theta \, dx = \frac{1}{\theta}(-\theta)\int_0^\infty x^2 de^{-x/\theta} =$$

$$= -\left[x^2 e^{-x/\theta}\Big|_0^{+\infty} - \int_0^\infty 2x e^{-x/\theta} dx\right] = 2\int_0^\infty x e^{-x/\theta} dx =$$

$$= 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

$$D\tilde{\theta}_1 = \frac{1}{n}\theta^2$$

$$D\tilde{\tilde{\theta}}_2 = D\left(\frac{6}{5}X_{(2)}\right) = \frac{36}{25} D X_{(2)}$$

$$M X_{(2)}^2 = \int_0^\infty x^2 \cdot \frac{6}{\theta} e^{-\frac{2x}{\theta}}\left(1 - e^{-x/\theta}\right) dx =$$

$$= \frac{6}{\theta}\left[\int_0^\infty x^2 e^{-\frac{2x}{\theta}} dx - \int_0^\infty x^2 e^{-3x/\theta} dx\right] =$$

$$= \frac{6}{\theta}\left[-\frac{\theta}{2}\int_0^\infty x^2 de^{-2x/\theta} + \frac{\theta}{3}\int_0^\infty x^2 de^{-3x/\theta}\right] =$$

$$= \frac{6}{\theta}\left[-\frac{\theta}{2}\left(x^2 e^{-2x/\theta}\Big|_0^\infty - \int_0^\infty 2x e^{-2x/\theta} dx\right) + \frac{\theta}{3}\left(x^2 e^{-3x/\theta}\Big|_0^\infty - \int_0^\infty 2x e^{-3x/\theta} dx\right)\right]$$

$$= \frac{6}{\theta}\left[\theta\int_0^\infty x e^{-2x/\theta} dx - \frac{2}{3}\theta\int_0^\infty x e^{-3x/\theta} dx\right] =$$

$$= \frac{6}{\theta}\left[\theta \cdot \frac{\theta^2}{4} - \frac{2}{3}\theta \cdot \frac{\theta^2}{9}\right] = \frac{6}{\theta}\left[\frac{\theta^3}{4} - \frac{2\theta^3}{27}\right] \ominus$$

$$= \frac{3}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{27\theta^2 - 8\theta^2}{18} = \frac{19}{18}\theta^2$$

$$D(X_{(2)}) = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$$

$$\Rightarrow D\tilde{\tilde{\theta}}_2 = \frac{36}{25} \cdot \frac{13}{36}\theta^2 = \frac{13}{25}\theta^2$$

$$D\hat{\theta}_1 = \frac{1}{3}\theta^2 \vee D\tilde{\tilde{\theta}}_2 = \frac{13}{25}\theta^2 \quad \left(\frac{25}{75} \vee \frac{39}{75}\right)$$

$$\forall \theta > 0 \quad D\hat{\theta}_1 \leq D\tilde{\tilde{\theta}}_2$$

$$\exists \theta > 0 : D\hat{\theta}_1 < D\tilde{\tilde{\theta}}_2 \Rightarrow \hat{\theta}_1 \text{ эффективнее } \tilde{\tilde{\theta}}_2$$

c) Цензурим модел на регулярност:

$$F(x,\theta) = \begin{cases} 1 - e^{-x/\theta}, & x > 0 \\ 0, & x \le 0 \end{cases}$$ - непр. диф. по θ на

$(0; +\infty)$

$$\int\limits_{A(\theta)} \frac{\partial p(x,\theta)}{\partial \theta} dx = 0 \quad A(\theta) = (-\infty; +\infty)$$

$$\int\limits_{-\infty}^{\infty} = \underbrace{\int\limits_{-\infty}^{0}}_{=0,\ \text{т.к. } \frac{\partial(0)}{\partial\theta} = 0} + \int\limits_{0}^{\infty} = \int\limits_{0}^{\infty} \frac{\partial}{\partial\theta}\left(\frac{1}{\theta} e^{-x/\theta}\right) =$$

$$= \left(-\frac{1}{\theta^2} e^{-x/\theta} + \frac{1}{\theta} e^{-x/\theta} \cdot \frac{x}{\theta^2}\right)$$

$$\int\limits_{0}^{\infty}\left(-\frac{1}{\theta^2} e^{-x/\theta}\right) dx + \int\limits_{0}^{\infty} \frac{1}{\theta^3} x e^{-x/\theta} dx = -\frac{\theta}{\theta^2} + \frac{\theta^2}{\theta^3} =$$

$$= -\frac{1}{\theta} + \frac{1}{\theta} = 0$$

$$I(\theta) = \int\limits_{-\infty}^{\infty} \left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 p(x,\theta) dx = \int\limits_{-\infty}^{\infty} \left(\frac{\partial \ln p}{\partial\theta}\right)^2 dF(x)$$

$$= \left\{ \begin{array}{l} \text{на } (0;+\infty): \ \ln \frac{1}{\theta} e^{-x/\theta} = -\ln\theta - x/\theta \\ \frac{\partial \ln p}{\partial\theta} = -\frac{1}{\theta} + \frac{x}{\theta^2}, \ \text{на } (-\infty;0): \int\limits_{-\infty} = 0, p = 0 \end{array} \right\} \textcircled{=}$$

плотности отлична от нуля

$$\textcircled{=} \int\limits_{0}^{\infty} \left(-\frac{1}{\theta} + \frac{x}{\theta^2}\right)^2 \frac{1}{\theta} e^{-x/\theta} dx = \cancel{\frac{1}{\theta^2}\theta + \frac{1}{\theta^3}} =$$

$$= \int\limits_{0}^{\infty} \left(\frac{x^2}{\theta^4} - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right) \frac{1}{\theta} e^{-x/\theta} dx = \frac{1}{\theta^5} \int\limits_{0}^{\infty} x^2 e^{-x/\theta} dx -$$

$$- \frac{2}{\theta^4} \int\limits_{0}^{\infty} x e^{-x/\theta} dx + \frac{1}{\theta^3} \int\limits_{0}^{\infty} e^{-x/\theta} dx =$$

$$= \frac{1}{\theta^5} \cdot -\theta\left[x^2 e^{-x/\theta}\Big|_0^{\infty} - 2\int\limits_{0}^{\infty} x e^{-x/\theta} dx\right] - \frac{2}{\theta^4} \cdot \theta^2 + \frac{1}{\theta^3} \cdot \theta =$$

$$\frac{2\theta^3}{\theta^5} - \frac{2\theta^2}{\theta^4} + \frac{\theta}{\theta^3} = \frac{1}{\theta^2}$$ – непр на $(0; +\infty)$

и $I(\theta) = \frac{1}{\theta^2} > 0 \quad \forall \theta > 0 \Rightarrow$

$\Rightarrow$ модель явл. регулярной

Рассм. $\hat{\theta}_1 = \overline{X} = f(\overrightarrow{X_n})$

$\hat{\theta}_1$ – несмещённая, $D\hat{\theta}_1 = \frac{1}{3}\theta^2$ – огранич. на $\forall$ компакте из $(0; +\infty)$ (при этом модель – регулярна) $\Rightarrow$ оценка явл. регулярной по доп. условию $\Rightarrow$ справедливо нер-во Крамера – Рао:

$$D\hat{\theta}_1 \geq \frac{1}{n \cdot I(\theta)} = \frac{1}{n \cdot \frac{1}{\theta^2}} = \frac{\theta^2}{n} \quad \forall \theta > 0, \; n = 3$$

Но $D\hat{\theta}_1 = \frac{1}{3}\theta^2$, $D\hat{\theta}_1 \geq \frac{\theta^2}{3} \Rightarrow$

$\Rightarrow$ оценка $\hat{\theta}_1$ явл. <u>эффективной</u>.

Рассм. $\tilde{\tilde{\theta}}_2 = \frac{6}{5} X_{(2)}$.

$\tilde{\tilde{\theta}}_2$ – несмещенная, $D\tilde{\tilde{\theta}}_2 = \frac{13}{25}\theta^2$ – огранич. на $\forall$ компакте из $(0; +\infty)$, модел. – регулярная $\Rightarrow$ оценка явл. регулярной $\Rightarrow$

$\Rightarrow$ по нер-ву Крамера – Рао:

$$D\tilde{\tilde{\theta}}_2 \geq \frac{1}{nI(\theta)} = \frac{\theta^2}{3} \quad \forall \theta > 0, \text{ но}$$

$$D\tilde{\tilde{\theta}}_2 = \frac{13}{25}\theta^2 > \frac{\theta^2}{3},$$ а нижняя грань достигается $\left(D\hat{\theta}_1 = \frac{\theta^2}{3}\right) \Rightarrow \tilde{\tilde{\theta}}_2$ не явл. эффективной.